

Comparative Analysis of Sequential Models for Mobile Network Traffic Forecasting

Formative Assignment 1 - Final Report

Tapiwanashe Gift Marufu

Bachelor of Software Engineering (Machine Learning)

African Leadership University

September 2026

Research Question

How do different sequential models compare for one-step-ahead mobile network traffic forecasting, and how does their performance vary across geographical areas with different traffic characteristics?

Abstract

Accurate short-term mobile network traffic forecasts can support capacity planning, load balancing, and resource allocation. This study compares a statistical Fourier-seasonal ARIMA model, a Long Short-Term Memory (LSTM) network, and a Transformer against a persistence baseline for one-step-ahead Internet-traffic forecasting at the original 10-minute resolution of the Telecom Italia Milan dataset. The raw data were processed incrementally because of their size, aggregated to one Internet-traffic observation per geographical square and timestamp, and analysed for temporal dependence, seasonality, stationarity, and spatial heterogeneity. Three complete grid squares were selected to represent low/stable, medium/moderate, and high/variable traffic profiles. A chronological 70/15/15 train-validation-test protocol was used, with model selection confined to the training and validation periods. On the held-out test set, the LSTM produced the lowest MAE, RMSE, and sMAPE in all three selected areas. Relative to persistence, its RMSE improved by 9.1%, 22.2%, and 21.6% for the low/stable, medium/moderate, and high/variable profiles respectively. Fourier-seasonal ARIMA was competitive in the low and medium profiles but slightly underperformed persistence in the high/variable area. The Transformer improved on persistence in the medium and high profiles but deteriorated in the low/stable area. Failure analysis showed that all approaches struggled during abrupt 10-minute changes and tended to underpredict high-traffic peaks. The results indicate that forecasting effectiveness depends on both model design and local traffic characteristics, with the evaluated LSTM configuration showing the most consistent relative performance across the selected areas.

Keywords: mobile network traffic, time-series forecasting, LSTM, Transformer, ARIMA, Fourier seasonality, Telecom Italia.

1. Introduction

Mobile networks exhibit strong temporal variation caused by recurring daily activity patterns, differences between geographical areas, and irregular bursts of demand. Forecasting this traffic is relevant to operational tasks such as resource allocation, network slicing, and load balancing. Statistical forecasting remains attractive because of its interpretability and relatively small parameter count, while recurrent and attention-based neural networks provide flexible alternatives for learning nonlinear sequential relationships.

This study treats the assignment as a comparative empirical investigation rather than a single-model prediction exercise. The central question is whether different sequential forecasting approaches behave consistently across areas with different traffic characteristics. To answer it, the study preserves the original 10-minute sampling interval, uses a common one-step-ahead information set, evaluates all models on identical chronological partitions, and compares performance against a persistence baseline.

The main contributions are: (1) a memory-efficient pipeline for processing the Milan telecommunications data; (2) time-series diagnostics covering seasonality, stationarity, autocorrelation and decomposition; (3) controlled validation experiments for a statistical model, LSTM and Transformer; (4) cross-area evaluation on three contrasting traffic profiles; and (5) failure analysis focused on residual bias, peak traffic and abrupt traffic changes.

2. Related Work

The dataset used in this work originates from the Telecom Italia Big Data Challenge and was documented by Barlacchi et al. as part of a multi-source urban dataset for Milan and Trentino [1]. Its fine spatial and temporal granularity makes it suitable for studying both temporal regularity and geographical heterogeneity.

Classical seasonal time-series models remain relevant to mobile-network forecasting. Kochetkova et al. applied SARIMA and Holt-Winters methods to mobile traffic and reported that both could provide useful short-term forecasts, with relative performance depending on the traffic direction [2]. This motivated inclusion of a statistical seasonal benchmark in the present study. However, an explicit SARIMA model with a 144-step daily seasonal period exceeded the available runtime memory at 10-minute resolution. Rather than changing the forecasting task, the statistical model was adapted to combine Fourier terms for daily seasonality with low-order ARIMA errors.

LSTM was introduced by Hochreiter and Schmidhuber to address difficulties in learning long-range dependencies with conventional recurrent networks [3]. Its gated memory mechanism provides a natural basis for modelling sequential traffic observations. The Transformer, introduced by Vaswani et al., replaces recurrence with attention and enables direct interaction between positions in an input sequence [4]. Subsequent time-series work such as Informer has shown the potential of attention-based models for long-sequence forecasting while also highlighting the computational challenges of standard self-attention [5]. Temporal Fusion Transformers further demonstrate how attention can be combined with recurrent processing and feature selection for forecasting [6].

These studies informed the comparative design used here: a statistical seasonal model tests whether explicit temporal structure is sufficient; the LSTM tests recurrent sequence learning; and a compact Transformer tests attention-based temporal modelling. The objective is not to reproduce the specialised architectures in prior work, but to compare representative modelling families under the same one-step-ahead protocol.

3. Dataset and Data Preparation

3.1 Dataset

The analysis uses 62 daily text files from the Milan telecommunications dataset, covering 1 November 2013 through 1 January 2014. The raw schema contains square identifier, Unix timestamp, country code, SMS-in, SMS-out, call-in, call-out and Internet activity. Multiple country-code rows can occur for the same square and timestamp, so Internet activity is aggregated across country codes.

3.2 Memory-Efficient Processing

A representative daily file contained 4,842,625 rows and required 295.57 MB when loaded with default Pandas numeric types. Only square_id, timestamp, country_code and internet were required for the forecasting task. The identifiers were stored as 16-bit integers, Internet activity as float32, and timestamps as int64. After geographical aggregation, the representative daily file occupied 19.23 MB, a 93.5% reduction relative to the unoptimized load. The complete dataset was processed incrementally rather than loading all raw files simultaneously.

Aggregation used sum(min_count=1) so that groups containing only missing Internet observations remained missing instead of being converted to zero. The complete aggregated dataset contained 89,245,318 square-timestamp observations, 10,000 geographical areas and 8,928 distinct 10-minute timestamps. Of the 10,000 areas, 9,902 had complete coverage and were eligible for modelling without global imputation.

3.3 Temporal Conversion and Missingness

Unix timestamps were converted to timezone-aware datetimes using Europe/Rome local time. The resulting sequence was verified to follow a 10-minute interval. Missing values were investigated rather

than automatically filled. Areas with complete coverage were selected for the final forecasting experiments, preserving the observed values and avoiding imputation-driven artefacts.

4. Exploratory and Time-Series Analysis

Exploratory analysis revealed substantial spatial heterogeneity. Total Internet activity was strongly right-skewed across geographical squares, indicating that a city-wide average would conceal important local differences.

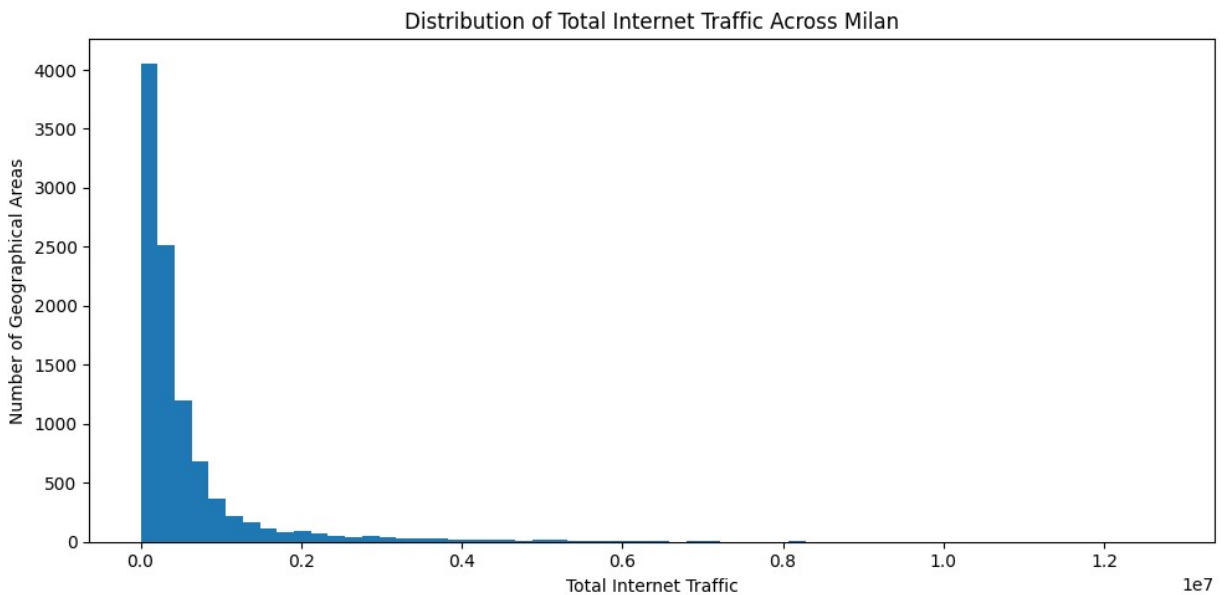


Figure 1. Distribution of Internet activity across Milan geographical squares.

The aggregated city series displayed a pronounced intraday pattern: traffic was lowest in the early morning, increased from approximately 06:00, and remained elevated through the daytime and evening. The data contain 144 ten-minute observations per day and 1,008 per week.

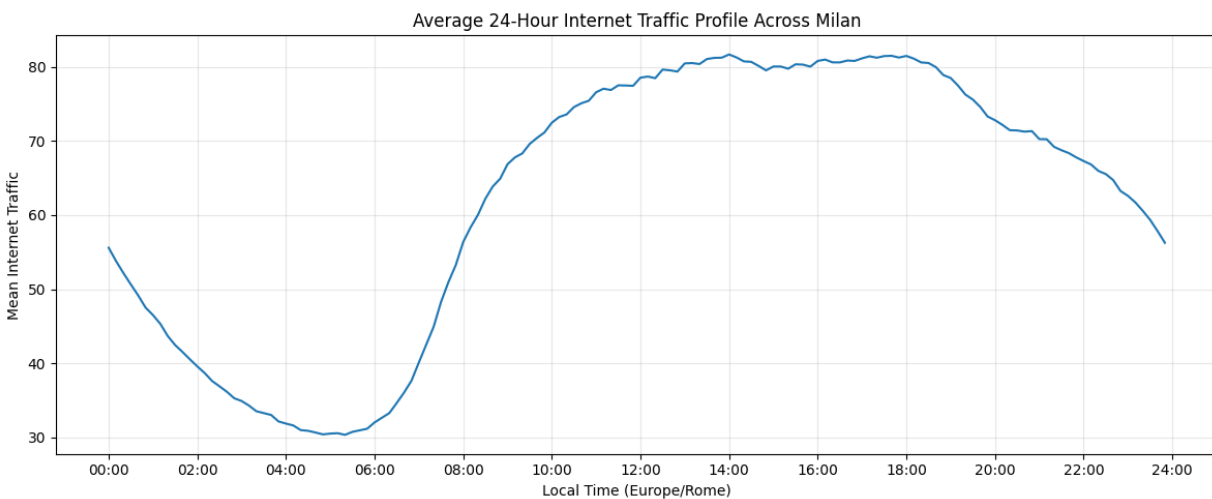


Figure 2. Average intraday Internet-traffic pattern.

Autocorrelation confirmed strong persistence and recurring daily and weekly structure. The city-level ACF was approximately 0.997 at one 10-minute lag, 0.958 at one hour, 0.920 at one day, and 0.802 at one week. The oscillating ACF pattern around daily multiples supports explicit treatment of seasonality.

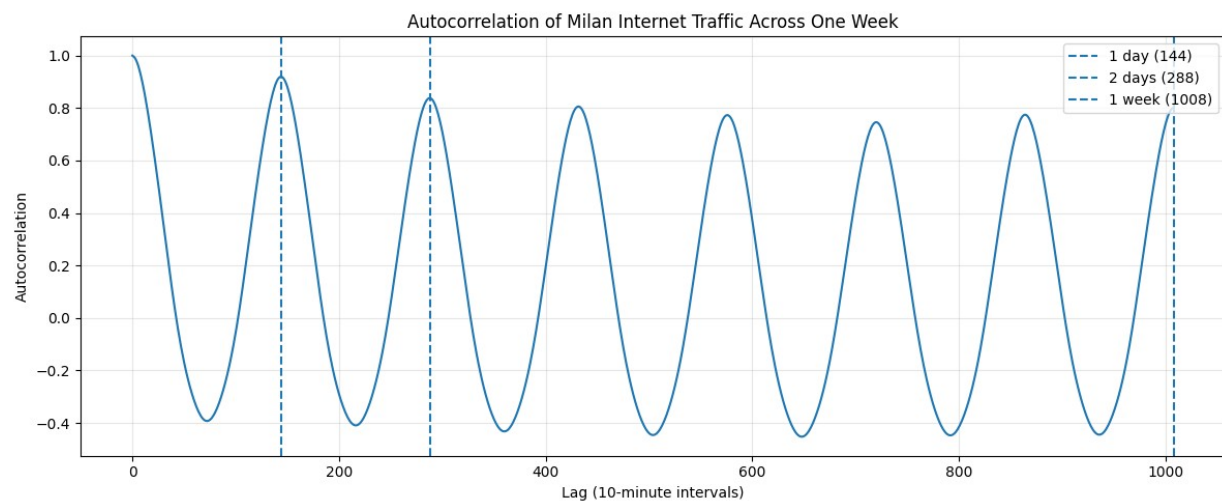


Figure 3. Autocorrelation structure of the city-level Internet-traffic series.

Stationarity tests produced a mixed result: the Augmented Dickey-Fuller test rejected a unit root, whereas the KPSS test rejected level stationarity. This is consistent with a series that contains strong deterministic seasonality and changing longer-term levels. First differencing sharply reduced immediate persistence but retained daily dependence, while daily seasonal differencing largely removed the lag-144 correlation. STL decomposition with period 144 estimated seasonal strength of 0.893 and trend strength of 0.718, confirming that both recurring daily structure and slower variation are important.

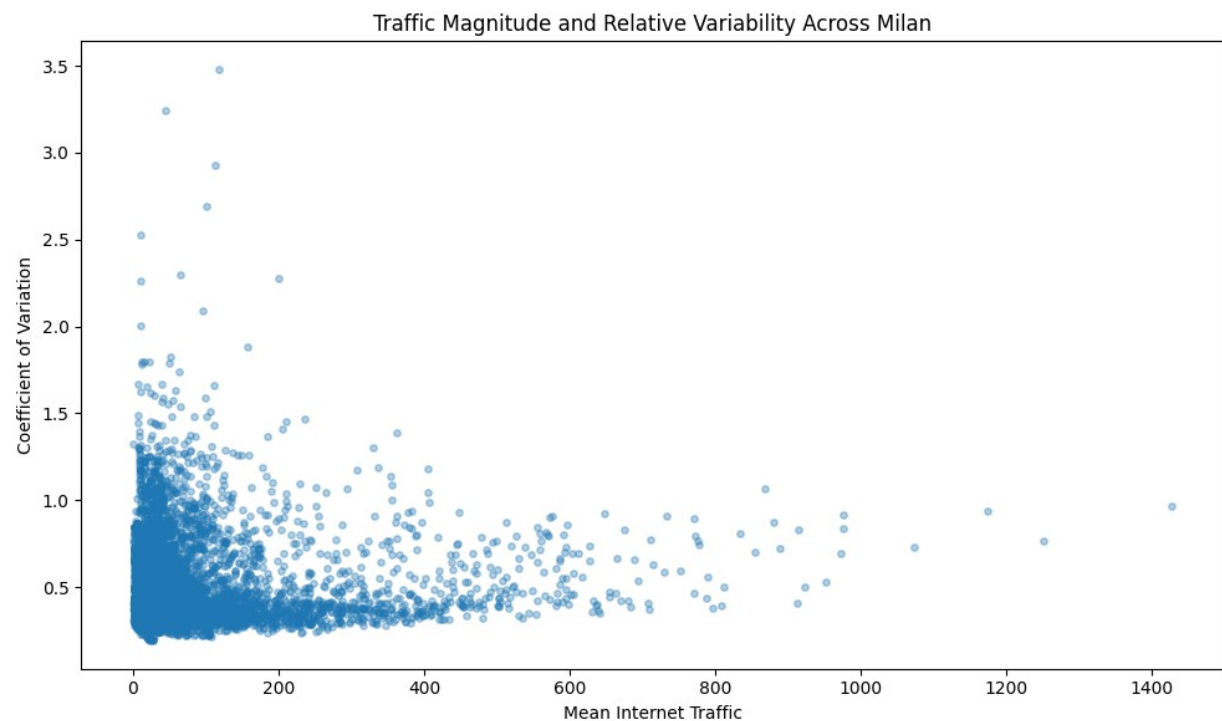


Figure 4. STL decomposition using a 144-step daily period.

5. Selection of Contrasting Geographical Areas

To make detailed experimentation computationally feasible while retaining geographical diversity, three complete grid squares were selected using mean traffic and coefficient of variation (CV). Candidate areas

were percentile-ranked and matched to target profiles: approximately the 25th percentile in mean and CV for low/stable traffic, the 50th percentile for medium/moderate traffic, and the 90th percentile for high/variable traffic.

Profile	Square	Mean	Std. Dev.	CV	Observations
Low/stable	2404	13.04	4.36	0.334	8,928
Medium/moderate	6433	30.94	12.17	0.393	8,928
High/variable	5243	148.07	113.55	0.767	8,928

The labels describe statistical traffic behaviour only; no land-use interpretation is inferred. Square 5243 also exhibits a pronounced level change around early December, making it useful for examining model sensitivity to changing traffic regimes.

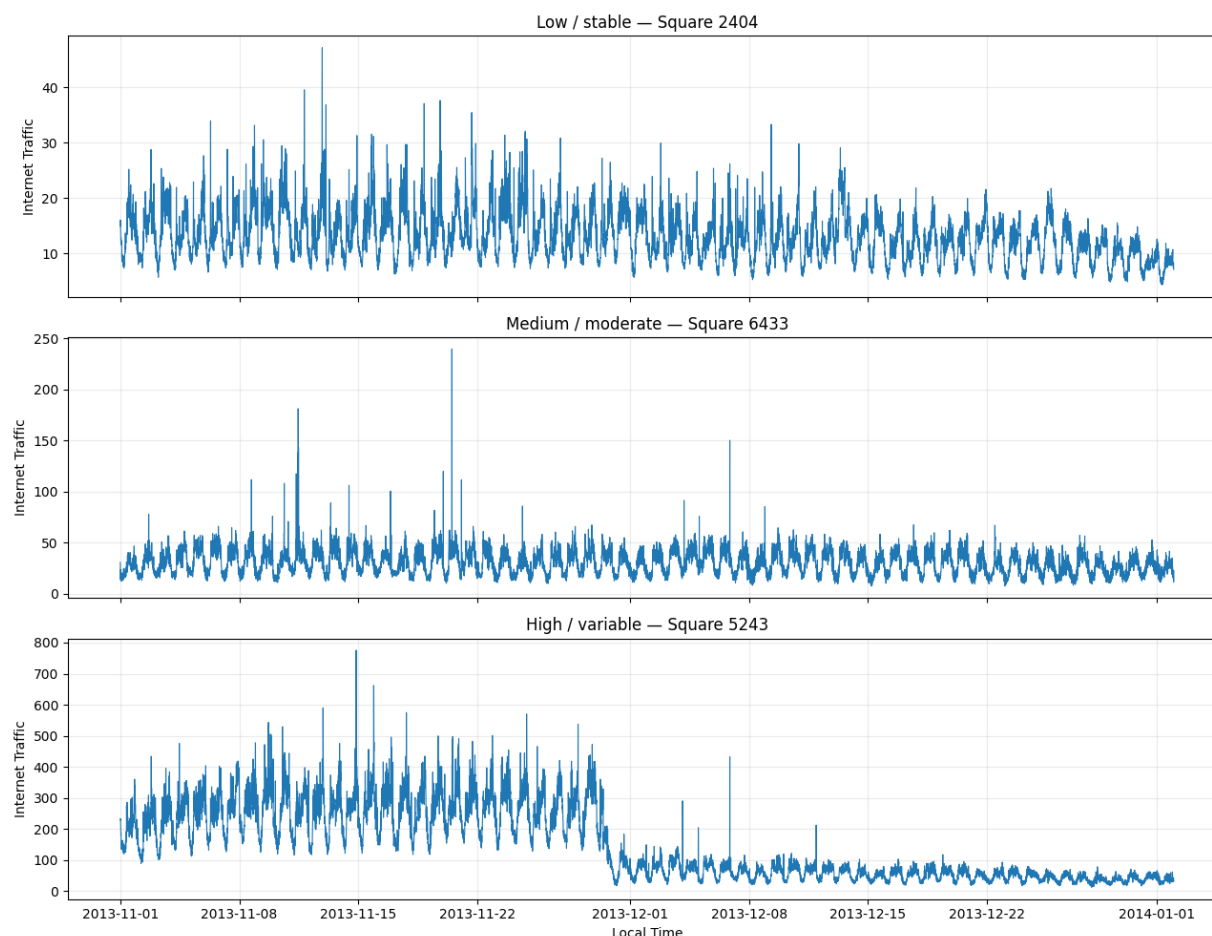


Figure 5. Time-series comparison of the three selected traffic profiles.

6. Methodology

6.1 Forecasting Protocol

The forecasting horizon is one step ahead, corresponding to 10 minutes. Each area contains 8,928 observations. Data were split chronologically into 70% training, 15% validation and 15% test: 6,249 training observations (1 Nov to 14 Dec 09:20), 1,339 validation observations (14 Dec 09:30 to 23 Dec 16:30), and 1,340 held-out test observations (23 Dec 16:40 to 1 Jan 23:50). The test targets were not

used for model fitting or hyperparameter selection. After configuration selection, each final model was refitted on the combined training and validation period and evaluated once on the test period.

For one-step-ahead evaluation, each prediction may use all observations available up to the immediately preceding timestamp. This means that previous actual test observations can enter the historical input window for later test predictions; no future target is used. This information set is shared by persistence, Fourier-seasonal ARIMA, LSTM and Transformer.

6.2 Evaluation Metrics

Performance was measured using mean absolute error (MAE), root mean squared error (RMSE), and symmetric mean absolute percentage error (sMAPE). RMSE provides stronger sensitivity to large errors, MAE gives an interpretable average absolute deviation, and sMAPE expresses relative error while avoiding division solely by the actual value. A persistence forecast, which predicts the next value using the most recently observed value, serves as the baseline.

6.3 Fourier-Seasonal ARIMA

A full SARIMA(1,0,1)(1,0,0,144) specification was attempted at the original 10-minute resolution but exceeded the available Colab memory before fitting completed. The study therefore retained the original resolution and represented the 144-step daily cycle using Fourier sine/cosine terms. Low-order ARIMA dynamics were then fitted with the Fourier terms as exogenous regressors. Four statistical configurations were compared on the low/stable development area. The selected specification was ARIMA(1,1,1) with two Fourier harmonics ($K=2$), based on validation performance and computational efficiency.

6.4 LSTM

The LSTM used StandardScaler parameters fitted only on the training series during model selection. The baseline architecture consisted of a 144-step lookback window, one 32-unit LSTM layer and a Dense output layer, trained with Adam and mean squared error. Experiments changed one major factor at a time: hidden units (32 to 64), lookback (144 to 72), and learning rate (0.001 to 0.0005). Early stopping with patience five restored the best validation weights. The original 32-unit, 144-step, 0.001-learning-rate configuration was selected.

6.5 Transformer

The Transformer projected each scalar observation into a learned feature representation, added learned positional embeddings, and passed the sequence through one encoder block containing multi-head self-attention, residual connections, layer normalization and a feed-forward network. Experiments varied sequence aggregation, model dimension, feed-forward dimension, learning rate and lookback length. The selected configuration used a 72-step lookback, $d_{\text{model}}=16$, two attention heads, feed-forward dimension 32, one encoder block, dropout 0.1, final-position aggregation and learning rate 0.0005.

6.6 Experimental Control

Hyperparameters were chosen from validation performance only. The same architecture settings were then applied across the three profiles, while final neural-network training duration used the best epoch previously identified for each area's validation run. No configuration was changed after viewing held-out test results.

6.7 Iterative Hyperparameter Tuning Evidence

Tuning followed an iterative process: each experiment changed a targeted parameter, validation performance was examined, and the next adjustment was justified by the observed result. The table below summarizes the main development-area experiments. Test data were not used in these decisions.

Model	Exp.	Change / setting	Validation result	Decision / next step
Fourier-ARIMA	1	(1,0,1), K=2	RMSE 11.5080	Poor vs persistence; add first differencing.
Fourier-ARIMA	2	(1,1,1), K=2	RMSE 9.5456	Best statistical validation result; retained.
Fourier-ARIMA	3	(3,1,1), K=2	RMSE 9.6043	Extra AR terms did not improve RMSE.
Fourier-ARIMA	4	(1,1,1), K=4	RMSE 9.5812	More harmonics added cost without improvement.
LSTM	1	144 lookback, 32 units, LR=.001	RMSE 1.3177	Strongest LSTM validation result; baseline retained.
LSTM	2	64 units	RMSE 1.3250	More capacity did not improve validation.
LSTM	3	72 lookback	RMSE 1.3217	Shorter context slightly worse.
LSTM	4	LR=.0005	RMSE 1.3253	Lower learning rate did not improve validation.
Transformer	1	144, d=32, heads=2, FF=64, LR=.001	RMSE 1.8982	Too large/weak; simplify architecture.
Transformer	2	Global-average pooling	RMSE 1.9847	Pooling worsened validation.
Transformer	3	d=16, FF=32	RMSE 1.6200	Smaller model improved validation.
Transformer	4	LR=.0005	RMSE 1.5029	Lower LR improved stability.
Transformer	5	72 lookback	RMSE 1.4554	Best Transformer validation result; selected.

7. Results and Discussion

7.1 Validation Findings

Validation experiments showed clear differences by model and profile. The selected LSTM reduced RMSE relative to persistence in all three areas: approximately 11.4% in the low/stable area, 22.9% in the medium/moderate area and 17.7% in the high/variable area. The selected Transformer also improved on persistence during validation, but by smaller margins of about 2.2%, 10.6% and 6.3% respectively. Fourier-seasonal ARIMA was highly profile-dependent: it performed very poorly in the low/stable validation window, improved RMSE by 20.7% in the medium area, and by only 2.1% in the high/variable area.

7.2 Held-Out Test Performance

Profile	Model	MAE	RMSE	sMAPE (%)
Low/stable	Persistence	0.9456	1.2387	9.3011
Low/stable	Fourier-ARIMA	0.8979	1.1630	8.9053
Low/stable	LSTM	0.8732	1.1264	8.6529
Low/stable	Transformer	1.2024	1.4794	12.5616
Medium/moderate	Persistence	5.3005	6.6660	21.6372

Medium/moderate	Fourier-ARIMA	4.1276	5.2465	16.7736
Medium/moderate	LSTM	4.0457	5.1868	16.1416
Medium/moderate	Transformer	4.9909	6.3359	19.8070
High/variable	Persistence	7.3329	9.3655	18.5031
High/variable	Fourier-ARIMA	7.6517	9.4290	20.9351
High/variable	LSTM	5.7516	7.3438	14.3716
High/variable	Transformer	6.9017	8.7723	17.1690

The held-out results answer the research question directly. The LSTM achieved the lowest MAE, RMSE and SMAPE in each selected area. Its RMSE reductions relative to persistence were 9.1% for low/stable traffic, 22.2% for medium/moderate traffic and 21.6% for high/variable traffic. This consistency is notable because the absolute scale and variability of the three series differ substantially.

Fourier-seasonal ARIMA was competitive in the low/stable and medium/moderate areas, reducing RMSE by 6.1% and 21.3% respectively. In the high/variable area, however, its RMSE was 0.7% worse than persistence. The result suggests that explicit daily seasonality and low-order autoregressive dynamics are useful where behaviour is comparatively regular, but are less robust to the high variability and regime behaviour observed in square 5243.

The compact Transformer produced mixed results. It was 19.4% worse than persistence in the low/stable test area, but improved RMSE by 5.0% and 6.3% in the medium and high profiles. The attention-based architecture therefore did not translate its additional modelling flexibility into uniformly stronger one-step forecasts under the selected data volume and tuning budget.

These findings are consistent with the methodological motivation from prior work rather than implying that one model family is universally superior. Classical seasonal models can be effective when recurring structure dominates [2], while LSTM was designed to retain sequential information through gated recurrence [3]. Transformer-based forecasting can capture longer-range interactions through attention [4]-[6], but the present experiment used a deliberately compact architecture, a short two-month dataset, and a one-step horizon. The weaker Transformer results should therefore be interpreted as evidence about this configuration and data regime, not as a general limitation of attention-based forecasting.

Computational behaviour also differed. Final Fourier-ARIMA fitting took approximately 1.6-3.4 seconds per area, but sequential rolling test forecasting required about 58-60 seconds. LSTM training ranged from about 24.6 to 92.7 seconds depending on the area, while batch prediction required less than 0.6 seconds. Transformer training ranged from about 18.9 to 29.8 seconds, with batch prediction around 1.4 seconds. These timings are not directly equivalent because the statistical model updates sequentially while the neural models predict preconstructed one-step windows in batches, but they illustrate an operational trade-off between fitting cost and inference strategy.

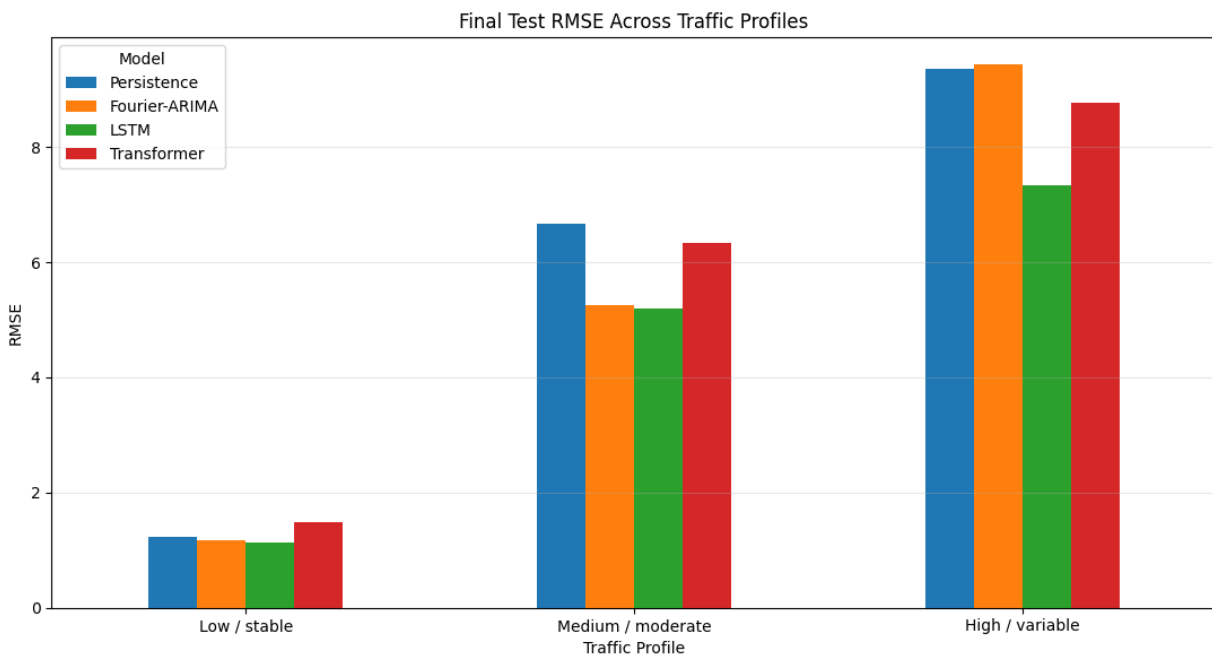


Figure 6. Held-out test RMSE across models and traffic profiles.

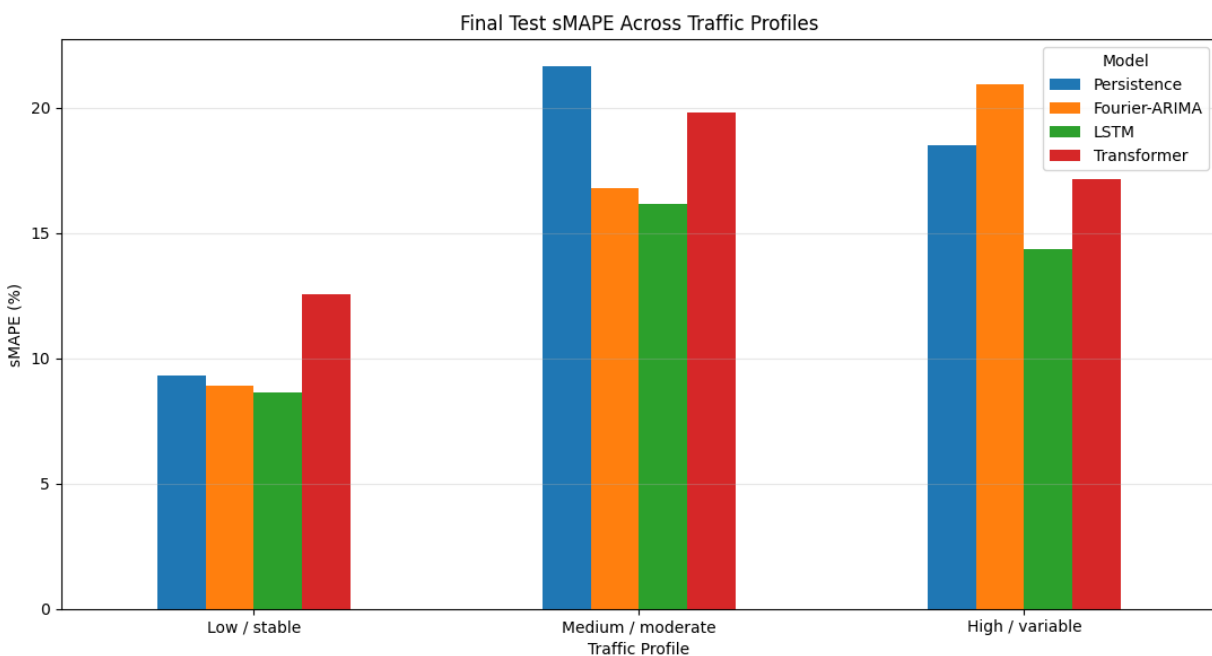


Figure 7. Held-out test sMAPE across models and traffic profiles.

7.3 Validation-to-Test Behaviour

Raw validation and test RMSE values should not be compared as if the periods had equal difficulty. Persistence itself improved from validation to test in all three areas, showing that the test period was generally easier for one-step forecasting. Final models were also refitted on the combined training and validation data, so they had access to more recent pre-test history.

Relative to persistence, the LSTM was the most stable configuration family: its RMSE improvement changed from 11.4% to 9.1% in the low/stable area, 22.9% to 22.2% in the medium area, and 17.7% to 21.6% in the high/variable area. The Transformer was less stable, especially in the low/stable area where

a 2.2% validation improvement became a 19.4% test deterioration. Fourier-seasonal ARIMA showed an extreme shift in the low/stable area, from severe validation underperformance to a 6.1% test improvement. This is treated as evidence of temporal sensitivity rather than exceptional generalization.

7.4 Failure Analysis

Residual analysis defined error as actual minus predicted. Persistence remained close to unbiased on average, whereas the learned models showed modest negative mean residuals, indicating a tendency to overpredict overall. This average behaviour changed during peaks: for the top 10% of actual traffic values, mean residuals were positive across models, indicating systematic underprediction of the highest-demand observations.

Errors also increased sharply during the largest 10% of absolute 10-minute traffic changes. In the high/variable area, abrupt-change RMSE was 20.31 for persistence, 14.97 for Fourier-seasonal ARIMA, 13.58 for LSTM and 17.92 for Transformer. The same pattern is visible around the largest high/variable test error on 28 December, where all models follow the broad trajectory but smooth rapid rises and falls.

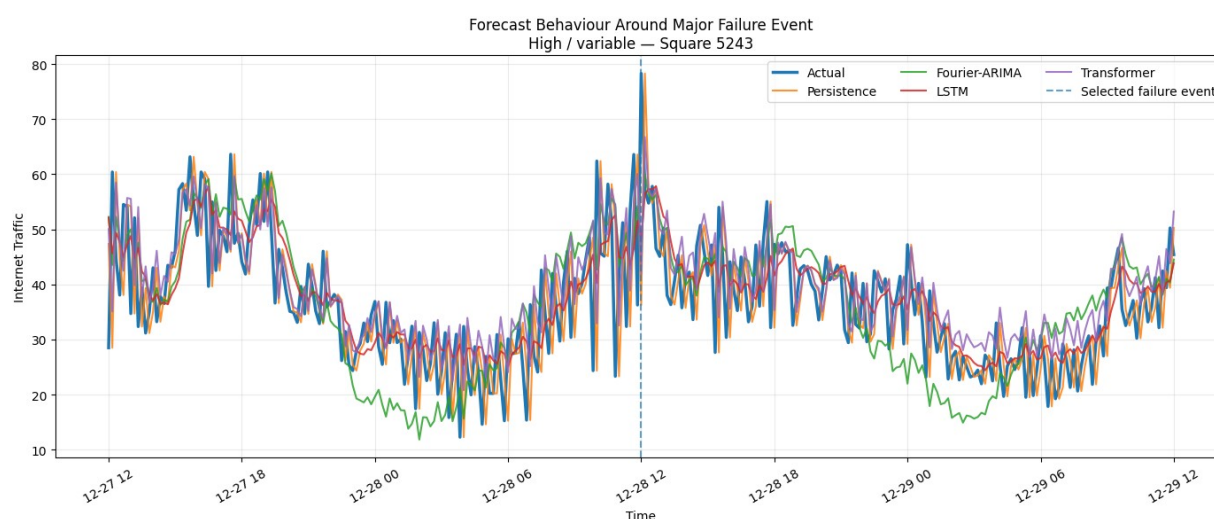


Figure 8. Forecast behaviour around a large error in the high/variable traffic area.

Peak and abrupt-change analyses also show why aggregate metrics alone are insufficient. For example, Fourier-seasonal ARIMA slightly outperformed the LSTM on low-area abrupt-change RMSE, while the LSTM had the best overall test metrics. In the high-area peak subset, Fourier-seasonal ARIMA had slightly lower RMSE than the LSTM despite being worse overall. Model behaviour is therefore condition-dependent even when one configuration has the strongest aggregate performance.

8. Limitations and Future Work

First, only three complete geographical areas were modelled in detail. They were deliberately selected to represent contrasting traffic profiles, but they cannot establish performance across all 10,000 grid squares. A larger evaluation using stratified sampling or clustering would strengthen geographical generalizability.

Second, area selection used full-period mean and coefficient-of-variation statistics before the forecasting split. This did not use test targets for model fitting or hyperparameter selection, but it means the profile labels were informed by the full observation period. A stricter future design would select profiles using training-period statistics only.

Third, the models are univariate and therefore cannot anticipate contextual events that are not encoded in recent Internet traffic. Calendar variables, neighbouring-cell traffic, events, weather or other exogenous signals could be evaluated as possible predictors of abrupt changes and high-demand periods.

Fourth, the study uses a single chronological validation and test window. The large low-area Fourier-ARIMA validation-to-test shift demonstrates the value of rolling-origin evaluation across multiple windows. Fifth, full seasonal SARIMA could not be fitted within the available memory at the original temporal resolution, so the statistical model is a Fourier-seasonal ARIMA approximation rather than explicit seasonal state-space SARIMA. Finally, neural hyperparameter search was targeted rather than exhaustive, and the high/variable LSTM's best validation epoch occurred close to the maximum training cap. These choices were retained after test evaluation to avoid post-test tuning.

Future work should therefore evaluate more geographical areas, use repeated rolling-origin splits, include longer multi-step horizons, and test spatial/contextual covariates. It would also be useful to compare specialised time-series Transformer variants and probabilistic forecasting methods under a larger computational budget.

9. Conclusion

This study compared persistence, Fourier-seasonal ARIMA, LSTM and Transformer forecasting for one-step-ahead Milan Internet traffic at 10-minute resolution across three contrasting geographical traffic profiles. The data showed strong daily seasonality, high short-lag dependence and substantial spatial heterogeneity, supporting the need for both temporal modelling and cross-area evaluation.

Among the evaluated configurations, the LSTM produced the lowest MAE, RMSE and sMAPE on the held-out test set in all three selected areas and maintained a positive RMSE improvement over persistence across both validation and test periods. Fourier-seasonal ARIMA was competitive in more regular profiles but sensitive to the evaluated period and high-variability traffic. The Transformer produced smaller and less consistent gains.

The failure analysis qualifies the aggregate results: all approaches had difficulty with abrupt transitions and tended to underpredict extreme peaks. The research question is therefore answered not by a universal claim that one architecture is always superior, but by evidence that forecasting effectiveness varies with local traffic characteristics and temporal conditions. Within this experiment, the LSTM showed the most consistent overall performance, while the statistical and attention-based models revealed useful strengths and weaknesses under specific traffic conditions.

References

- [1] G. Barlacchi et al., "A multi-source dataset of urban life in the city of Milan and the Province of Trentino," *Scientific Data*, vol. 2, Art. no. 150055, 2015, doi: 10.1038/sdata.2015.55.
- [2] I. Kochetkova, A. Kushchazli, S. Burtseva, and A. Gorshenin, "Short-Term Mobile Network Traffic Forecasting Using Seasonal ARIMA and Holt-Winters Models," *Future Internet*, vol. 15, no. 9, Art. no. 290, 2023, doi: 10.3390/fi15090290.
- [3] S. Hochreiter and J. Schmidhuber, "Long Short-Term Memory," *Neural Computation*, vol. 9, no. 8, pp. 1735-1780, 1997, doi: 10.1162/neco.1997.9.8.1735.
- [4] A. Vaswani et al., "Attention Is All You Need," in *Advances in Neural Information Processing Systems*, 2017, pp. 5998-6008.

- [5] H. Zhou et al., "Informer: Beyond Efficient Transformer for Long Sequence Time-Series Forecasting," in Proc. AAAI Conf. Artificial Intelligence, vol. 35, no. 12, pp. 11106-11115, 2021, doi: 10.1609/aaai.v35i12.17325.
- [6] B. Lim, S. O. Arik, N. Loeff, and T. Pfister, "Temporal Fusion Transformers for interpretable multi-horizon time series forecasting," International Journal of Forecasting, vol. 37, no. 4, pp. 1748-1764, 2021, doi: 10.1016/j.ijforecast.2021.03.012.

Submission Links

GitHub Repository: [Mobile Network Traffic Forecasting Repository](#)

Video Presentation: [Video Presentation](#)

Appendix A. Reproducibility Summary

The final GitHub repository should contain the refined notebook/source code, a README with setup and execution instructions, and a dependency/configuration file. The raw multi-gigabyte dataset should be referenced through its acquisition method rather than committed to the repository.

Component	Final setting
Temporal resolution	10 minutes
Forecast horizon	One step ahead (10 minutes)
Split	70% train / 15% validation / 15% test
Persistence	Previous observed value
Fourier-ARIMA	ARIMA(1,1,1), K=2, period=144
LSTM	Lookback 144, 32 units, LR 0.001, batch 32
Transformer	Lookback 72, d_model 16, 2 heads, FF 32, 1 block, dropout 0.1, LR 0.0005
Metrics	MAE, RMSE, SMAPE
Random seed	42

Appendix B. AI Use Disclosure

Generative AI tools were used during this project to support code development and debugging, improve the organisation and documentation of the notebook, refine explanations of experimental results, and assist with structuring and drafting the final report. All experiments were executed by the student, and the reported numerical results were obtained from the executed notebook. The student reviewed the generated material, evaluated the modelling decisions and results, and remains responsible for the methodology, interpretation, and conclusions presented in this work.